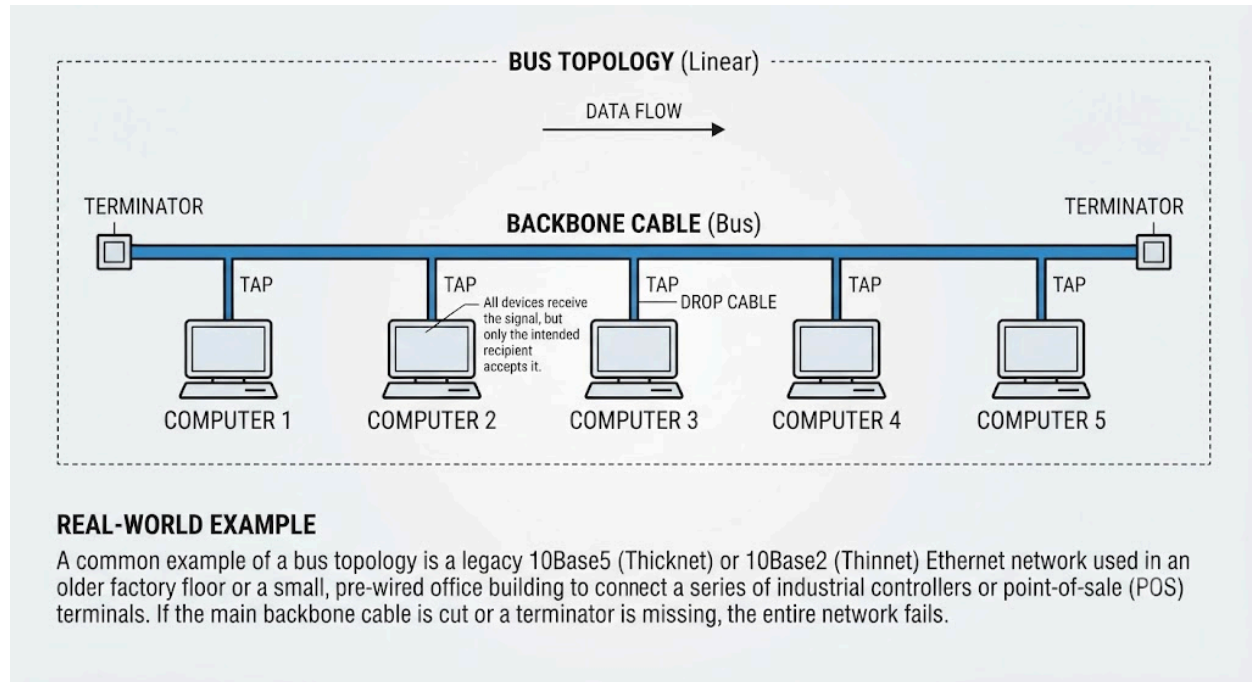


## DAY 2 NETWORK TOPOLOGIES THEORY

Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

ANS:



In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

ANS:

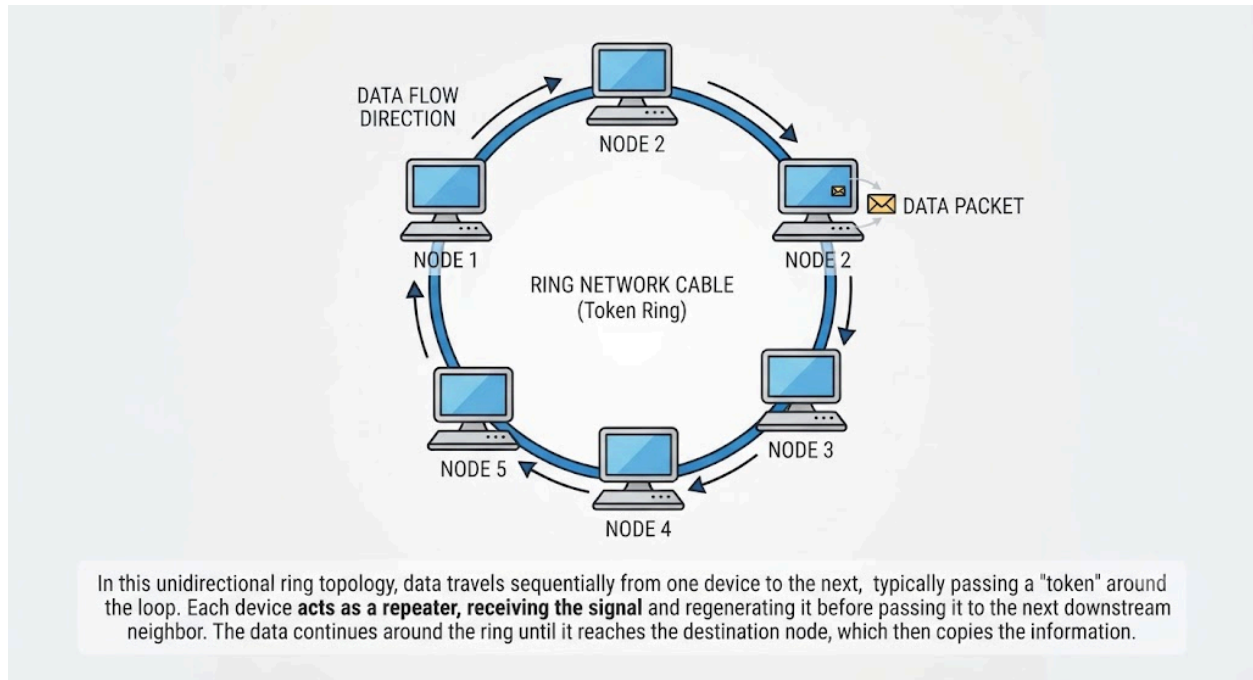
**Setup Steps:** Open Packet Tracer → Add 1 Switch and 4 PCs → Rename PCs (**WhatsApp**, **Zomato**, **Spotify**, **Flipkart**).

**Connection:** Use **Copper Straight-Through** cables to link each PC's Ethernet port to the switch's ports.

**Verification:** Green link lights confirm a successful star topology connection.

Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.  
  
**Hint:** Use arrows to show the direction of data flow.

ANS:



Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

ANS:

Here is a comparison of mesh and star topologies:

| Topology      | Advantages                                                                                                                                                                                                                                                      | Disadvantages                                                                                                                                                                                                                                |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Star Topology | <ul style="list-style-type: none"> <li>• <b>Easy Maintenance:</b> Adding or removing devices is simple without disrupting the rest of the network.</li> <li>• <b>Fault Isolation:</b> If one cable fails, only that specific device is disconnected.</li> </ul> | <ul style="list-style-type: none"> <li>• <b>Single Point of Failure:</b> If the central switch or hub fails, the entire network goes down.</li> <li>• <b>Cabling Cost:</b> Requires more cable than linear topologies like a bus.</li> </ul> |

|                      |                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      |                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                             |
| <b>Mesh Topology</b> | <ul style="list-style-type: none"><li>• <b>High Redundancy:</b> Multiple paths exist for data, ensuring uninterrupted communication if one link breaks.</li><li>• <b>Maximum Reliability &amp; Security:</b> Highly fault-tolerant and secure since data flows through dedicated paths.</li></ul> | <ul style="list-style-type: none"><li>• <b>High Cost &amp; Complexity:</b> Requires a massive amount of cabling and multiple ports per device, making it expensive.</li><li>• <b>Difficult Setup:</b> Configuration and management are complex due to the dense interconnections.</li></ul> |